

A SIMULATION TESTBED FOR EVALUATING MULTI-SENSOR DETECT-AND-AVOID PERCEPTION BEFORE FLIGHT TEST

Pedro Henrique Blatt Machado

Instituto Tecnológico de Aeronáutica (ITA), São José dos Campos, SP, BRAZIL

pedro.blatt@ita.br

André Kuroswiski

Instituto de Controle do Espaço Aéreo (ICEA), São José dos Campos, SP, BRAZIL

andre.kuroswiski@icea.gov.br

ABSTRACT

Flight-testing non-cooperative encounters is costly and cannot repeat an identical geometry across competing sensor suites, so the sensing envelope that determines detect-and-avoid warning time is rarely measured directly. We present a simulation testbed built on Cosys-AirSim that generates repeatable encounters and indexes every radar, LiDAR, and camera observation to the same frame-level ground-truth intruder range, with detector labels derived automatically from segmentation. A thirty-run campaign spanning approach geometry, weather, and lighting demonstrates the design questions the testbed answers before any flight: modality complementarity by range band, with vision peaking between twenty-five and one hundred meters while radar presence cueing spans all bands, lighting invariance, and whether one vision model generalizes across four camera configurations. Results are simulator-specific and false-alarm rates are not measured; we position the testbed as the first stage of a staged validation path whose next step is quantifying simulation-to-reality transfer in flight.

1 INTRODUCTION

Detect-and-avoid (DAA) alerting logic such as DAIDALUS is commonly evaluated assuming intruder state is already available (NASA WellClear Project 2020; RTCA Inc. 2020), leaving untested the sensing envelope that sets warning time against non-cooperative traffic without Automatic Dependent Surveillance–Broadcast (ADS-B). Sensor Uncertainty Mitigation expands Well Clear volumes under noisy measurements (Abramson, Narkawicz, and Muñoz 2020), but still needs modality-specific detection envelopes that flight test cannot supply at acceptable cost: an identical encounter cannot be repeated across competing sensor suites, and frame-level ground-truth range is unavailable simultaneously for radar, LiDAR, and camera. High-fidelity multi-modal simulation can hold the encounter fixed and vary only the sensor chain (Shah et al. 2018; Jansen et al. 2023). Contributions: (i) a Cosys-AirSim testbed with a frame-level ground-truth indexing contract; (ii) a factorial encounter design over geometry, weather, and lighting; (iii) a thirty-run demonstration of the pre-flight design questions the testbed answers; and (iv) a declared staged path from synthetic campaign to instrumented flight test.

2 SIMULATION TESTBED AND EXPERIMENT DESIGN

The testbed builds on Cosys-AirSim (Jansen et al. 2023) and is reusable for two reasons: detector labels are derived automatically from segmentation at zero annotation cost, and detections align by `(run, frame)` to telemetry `dist_xy_m`, which makes cross-modal comparison fair. Thirty runs vary approach azimuth, vertical separation, weather (clear, fog, rain, dust, snow), and lighting (day, dawn, dusk, night). Modalities: an FMCW radar return-presence gate (no clutter, no classification); high-density LiDAR (OS128) and sparse LiDAR (VLP-16) with geometric clustering (Ester et al. 1996); and four forward EO cameras—standard

(1280×960, 120°), medium (1280×960, 90°), narrow HD (1920×1080, 65°), and low-cost wide (640×480, 78°)—processed by a unified YOLOv8s (Jocher, Glenn and Chaurasia, Ayush and Qiu, Jing 2023) trained on pooled segmentation-derived labels. EO rates use any-camera-detects over 976 frames (30 runs); LiDAR and radar use 640 synchronized frames (29 runs).

3 DEMONSTRATION STUDY

The campaign answers three pre-flight design questions that flight test cannot isolate at this scale. Table 1: EO peaks mid-range (91.1% at 25–50 m; 100% at 50–100 m), matching or exceeding dense LiDAR there; the drop below 25 m is geometric and beyond 100 m angular size collapses. Radar stays at 100% in every band as an upper-bound presence signal under a no-clutter gate. Overall rates: radar 100% (640/640), dense LiDAR 63.1% (404/640), EO 49.7% (485/976), sparse LiDAR 16.1% (103/640). Mean first-detection ranges are 116.9 m (radar), 113.2 m (dense LiDAR), and 79.3 m (EO; not frame-aligned)—about 7.8 s, 7.5 s, and 5.3 s of warning at 15 m/s closure. Radar and LiDAR are lighting-invariant; EO varies from 39.5% (dawn) to 65.9% (night). Table 2: the unified model matches or slightly exceeds each per-lens specialist (mAP50 0.92–0.99); splits are small, so the low-cost score is not evidence that lower resolution is better.

Dist. (m)	EO	HD	Sp.	Rad.
0–10	55.6	77.6	29.9	100
10–25	44.3	77.1	32.1	100
25–50	91.1	67.7	16.5	100
50–100	100.0	55.4	11.2	100
100–150	2.0	58.5	0.0	100
150–250	0.0	6.7	0.0	100

Table 1: Detection rate (%). EO: 976/30 runs. LiDAR/Radar: 640/29. Radar is return-presence only.

Camera	FOV	<i>N</i>	Sp.	Un.
Narrow HD	65°	41	0.91	0.92
Medium	90°	54	0.96	0.97
Low-cost	78°	37	0.99	0.99
Pooled	—	170	0.96	0.97

Table 2: Per-camera mAP50 (Spec./Unif.). Standard forward is pooled only.

Figure 1 stresses the same pipeline across environments inside simulation: camera optics in an unseen urban scene (left pair) and clear day versus rain frames (right pair).

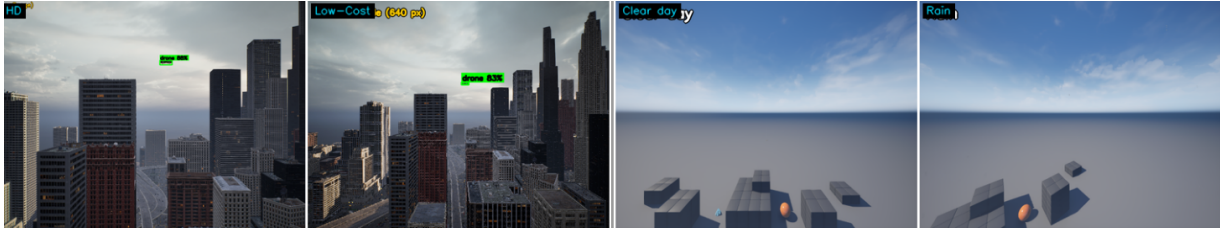


Figure 1: Left pair: Narrow HD vs Low-Cost (urban). Right pair: Clear day vs rain. Within-sim stress tests.

4 CONCLUSIONS AND PATH TO FLIGHT TEST

The testbed makes complementary, range-stratified sensing envelopes measurable before any aircraft leaves the ground: radar for early presence, dense LiDAR for geometry, and EO for mid-range classification with multi-lens generalization. Validation stages: (i) synthetic campaign, this work; (ii) cross-environment stress inside simulation (Figure 1); (iii) hardware-in-the-loop; and (iv) instrumented flight test quantifying the simulation-to-reality gap on these same metrics (Tobin et al. 2017). False-alarm baselines and lateral/overtaking geometries remain open on that roadmap; current scenarios are frontal-biased.

REFERENCES

- Abramson, M., A. J. Narkawicz, and C. A. Muñoz. 2020. “Applying Sensor Uncertainty Mitigation Schemes to Detect-and-Avoid Systems”. In *Proc. AIAA/IEEE Digital Avionics Systems Conference (DASC)*.
- Ester, M., H.-P. Kriegel, J. Sander, and X. Xu. 1996. “A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise”. In *Proc. 2nd Intl. Conf. on Knowledge Discovery and Data Mining (KDD)*, 226–231.
- Jansen, W., E. Verreycken, A. Schouten, D. Simon, C. Cornelis, and P. Hellinckx. 2023. “Cosys-AirSim: A Simulation Framework for Multi-Modal Sensing in UAV Applications”. In *Proc. International Conference on Unmanned Aircraft Systems (ICUAS)*, 1–8.
- Jocher, Glenn and Chaurasia, Ayush and Qiu, Jing 2023. “YOLO by Ultralytics”. <https://github.com/ultralytics/ultralytics>. accessed 6th August 2026.
- NASA WellClear Project 2020. “WellClear: Well-Clear Boundary Models for Integration of UAS in the NAS”. <http://nasa.github.io/WellClear/>. accessed 6th August 2026.
- RTCA Inc. 2020. “DO-365B: Minimum Operational Performance Standards (MOPS) for Detect and Avoid (DAA) Systems”. Technical report, RTCA Special Committee 228.
- Shah, S., D. Dey, C. Lovett, and A. Kapoor. 2018. “AirSim: High-Fidelity Visual and Physical Simulation for Autonomous Vehicles”. In *Field and Service Robotics*, 621–635.
- Tobin, J., R. Fong, A. Rav, J. Schneider, W. Zaremba, and P. Abbeel. 2017. “Domain Randomization for Transferring Deep